Exploring cooperation mechanisms via reinforcement learning in network common-pool resource games

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Abstract

Sustaining cooperation in resource-constrained populations requires allocation mechanisms that balance individual incentives, resource sustainability, and distributional fairness. This paper proposes a network common-pool resource game in which individuals are embedded in complex networks, participate in multiple overlapping local resource pools, and face endogenous resource constraints during strategy evolution. Within this framework, we first examine two representative allocation mechanisms, equal allocation and proportional allocation. The results show that equal allocation produces fair but inefficient outcomes by weakening contribution incentives, whereas proportional allocation can temporarily promote cooperation but amplifies accumulated advantages and leads to severe inequality. To overcome these limitations, we develop a graph neural network-based reinforcement learning framework in which a learned social planner allocates local pool resources without directly controlling individual strategies. Simulation results under four representative network topologies show that the learned planner sustains higher cooperation levels and average accumulated resources, and reduces inequality compared with the baselines. Furthermore, we interpret the learned policy and distill it into two simpler mechanisms: a resource-dependent mixture mechanism for regular networks and a degree-conditioned mixture mechanism for heterogeneous networks. These mechanisms reveal that effective allocation should adapt to both local resource states and structural positions, providing an interpretable route from reinforcement learning policy search to mechanism design in networked resource-sharing systems.

Keywords: Cooperation evolution, Deep reinforcement learning, Network common-pool resource game, Interpretable allocation mechanism

1. Introduction

Cooperation in human societies often relies on common resources that must be repeatedly allocated, used, replenished, and redistributed [1–5]. Governments allocate public funds to citizens and firms with the expectation of future taxes, innovation, or civic responsibility; organizations distribute resources to employees and teams whose effort sustains collective productivity; digital platforms provide visibility or shared infrastructure to users whose participation maintains the platform ecosystem; and ecological systems such as fisheries, forests, and irrigation networks require users to restrain short-term extraction to preserve future availability [6–13]. These situations all contain a common-pool resource problem: once resources are allocated, each recipient may either contribute back to the collective system or retain the benefits for private use. Although non-contribution can be individually advantageous in the short run, its spread gradually weakens resource regeneration, reduces future allocation capacity, and undermines long-term collective welfare. The central question in common-pool resource games is therefore how allocation mechanisms can sustain cooperation among self-interested individuals while preventing the depletion of the resource system from which they all benefit [14–17]. This question becomes particularly challenging in networked populations, where resources are

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not exchanged through a single global pool but through local and overlapping interaction structures. Individuals may participate in different numbers of local pools, occupy unequal structural positions, and face heterogeneous risks of resource depletion. As a result, an allocation rule that appears fair or efficient within a single pool may generate unintended inequality or cooperation collapse at the population level.

A large body of research has investigated mechanisms that promote cooperation in relative social dilemmas and networked populations. Classical examples include voluntary participation [18–23], reputation and punishment [24–29], prosocial exclusion [30, 31], network reciprocity [32–34], and related institutional or structural mechanisms [35, 36]. These mechanisms can align individual incentives with collective welfare by changing outside options, social information, sanctions, or interaction patterns. Recent work has further examined the tension between cooperation and fairness in the allocation of public goods on networks [37]. However, that study considers an edge-based public goods games and does not incorporate endogenous individual resource accumulation or resource-constrained participation. It therefore differs from the node-centred, overlapping common-pool resource system considered here, where each individual is also associated with a local resource pool and accumulated resources directly affect future cooperative capacity. Yet they leave open a distinct mechanism-design question: when a common resource has been generated, how should it be allocated back to individuals so that future cooperation remains possible? In common-pool resource systems, allocation is not merely a consequence of cooperation, but a driver of subsequent contribution capacity, resource regeneration, and inequality. This issue is especially challenging in networked populations, where resource exchange is local, and individuals occupy different structural positions. These considerations motivate the search for allocation rules that are not fixed globally, but adapt to both local resource conditions and network structure.

Recent advances in reinforcement learning have provided a computational route to explore cooperation-promoting strategies and mechanisms that are difficult to derive by hand. In evolutionary games, learning-based methods have been used to search large strategy spaces beyond classical heuristic rules, revealing adaptive strategies that can promote cooperation and improve social welfare in evolving populations [38–46]. These studies show that reinforcement learning can serve not only as a model of adaptive decision making, but also as a tool for discovering behavioural rules that would be hard to identify through analytical reasoning alone. More recently, this perspective has been extended from individual strategy learning to mechanism design, where a learning agent acts as an external social planner and searches for interventions that improve collective outcomes. For example, deep reinforcement learning has been used to scaffold cooperation in human groups by recommending changes to social connections [47, 48], and to design dynamic resource-allocation mechanisms that promote sustainable reciprocation in four individuals' common-pool resource problems [49]. These developments are part of the broader direction of deep mechanism design, in which deep reinforcement learning are used to discover socio-economic policies that are hard to specify analytically [50].

These studies have laid a solid foundation for exploring cooperation mechanisms within the context of game theory. And a more specific question can be raised for networked common-pool systems: can a resource-allocation mechanism sustain cooperation when resource exchange is local, overlapping, and structurally heterogeneous? In such systems, a rule that appears fair within a single pool may not be fair at the population level, because individuals with different network positions participate in different numbers of pools and face different risks of resource depletion. Likewise, a rule that rewards contribution may create strong incentives locally, but may also amplify early advantages and push resource-poor individuals into a persistent poverty trap. Therefore, the central difficulty is not simply whether equality or proportionality is preferable, but how a mechanism should adapt redistribution to both the current resource state and the structural position of the local pool.

To address this problem, we formulate a network common-pool resource game in which each node is both an individual and a local resource pool. Individuals choose whether to cooperate or defect through evolutionary dynamics, while cooperators contribute resources to the resource pools centered on themselves and their neighbours. The grown resources in each pool are then allocated back to the corresponding local group. Importantly, each individual has an endogenous resource stock, and individuals with insufficient resources are unable to cooperate. This setting explicitly couples strategy evolution, resource accumulation, local interaction structure, and allocation mechanisms. Because the consequences of an allocation rule are delayed and mediated by the evolving network state, manually designing an adaptive mechanism is difficult. We therefore train a graph neural network-based reinforcement learning agent as a social planner to explore mechanisms that promote cooperation. The planner does not directly determine individual strategies or rewire the network. Instead, it observes the evolving networked resource system and decides how each local pool should allocate resources among agents in its ego network. This allows the allocation mechanism to adapt to both local resource conditions and network structure while preserving the evolutionary nature of individual behaviour.

More importantly, we use the learned policy not only as a high-performing black-box mechanism, but also as a tool for mechanism discovery. By analysing and distilling the learned policy, we obtain two interpretable mixture mechanisms. On regular networks, the policy can be approximated by a resource-dependent mechanism whose allocation weights vary with the local resource state. On heterogeneous networks, resource information alone is insufficient, and a degree-conditioned mechanism is required. This reveals a structure-aware allocation principle: peripheral nodes require protection from poverty traps, intermediate-degree nodes require contribution-based incentives, and hub-like nodes require redistribution to prevent excessive resource concentration. These provide insights into the ways of redistributing resources in the current economic society to promote cooperation, efficiency, and fairness

2. Game Model

Understanding how cooperation emerges and is sustained in resource-constrained environments is a central problem in evolutionary game theory and socio-economic systems. Game Models such as the public goods game[8, 18, 26, 37] and the common-pool resource game[9, 16, 17, 49] have been proposed to describe and understand this problem. In many real-world scenarios, individuals are embedded in structured populations, where interactions are local, and resources are limited. Under such conditions, individuals' strategic decisions not only influence their payoffs but also determine their ability to participate in future interactions. Furthermore, under what principles or mechanisms, whether fairness-oriented equal distribution, contribution-based proportional distribution, or some other approach, should resources be redistributed to foster and sustain cooperation? To capture these coupled dynamics and problems, we propose a network common-pool resource game with endogenous resource constraints and allocation mechanisms.

2.1. Networked common-pool resource environment

We consider a population of N individuals, each with a resource pool structured by a social or economic network $G = (V, E)$, where $V = \{1, 2, \ldots, N\}$ is the set of individuals and E is the set of edges representing pairwise interactions. Each individual i possesses resources $R_i(t)$, with an initial resource of R_0. At each time step t, individual i selects a strategy $x_i(t) \in \{C, D\}$, where C represents cooperation and D represents defection.

If individual i with degree of d_i chooses cooperation, it will contribute a total resource of

$$e_i(t) = d_i + 1 + \alpha \max\{0, R_i(t) - (d_i + 1)\}, \tag{1}$$

uniformly to the resource pools of its ego network $N_i = \{j \in V \mid (i, j) \in E\}$, which is $e_i(t)/(d_i + 1)$, where α is the resource contribution ratio. However, if an individual's resource level is below the minimum requirement $d_i + 1$, the individual cannot cooperate; we refer to this constraint as a poverty trap. If individual i chooses defection, it will not contribute any resources, which means $e_i(t) = 0$.

2.2. Pool growth and baseline allocation mechanisms

The resource pool associated with individual i receives a total contribution of

$$P_i(t) = \sum_{j \in N_i \cup \{i\}} \frac{e_j(t)}{d_j + 1} \tag{2}$$

from the members of its ego network accordingly and then grows at rate r with maximum $P_{\max,i}(t)$. This reflects that resources, such as ecological stocks, public funds, or organizational resources, cannot expand without bound. The grown pool is

$$G_i(t) = \min\{(1 + r)P_i(t), P_{\max,i}(t)\}, \tag{3}$$

where $P_{\max,i}(t) = p_c\, N_{c,i}(t)\, \rho_{c,i}(t)$. Here, $p_c > 0$ is a scaling coefficient representing the basic productivity capacity per cooperator, $N_{c,i}(t) \geq 0$ denotes the number of cooperators in pool i at time t, and $\rho_{c,i}(t) \in [0, 1]$ is the ego-network cooperation fraction to reflect the efficiency of cooperation defined as $\rho_{c,i}(t) = \frac{N_{c,i}(t)}{d_i+1}$,

The resource $G_i(t)$ of the growth resource pool associated with individual i is allocated to the members of its ego network according to certain allocation mechanisms. These are two classic allocation mechanisms within the socioeconomic system: fairness-oriented equal distribution and contribution-based proportional distribution.

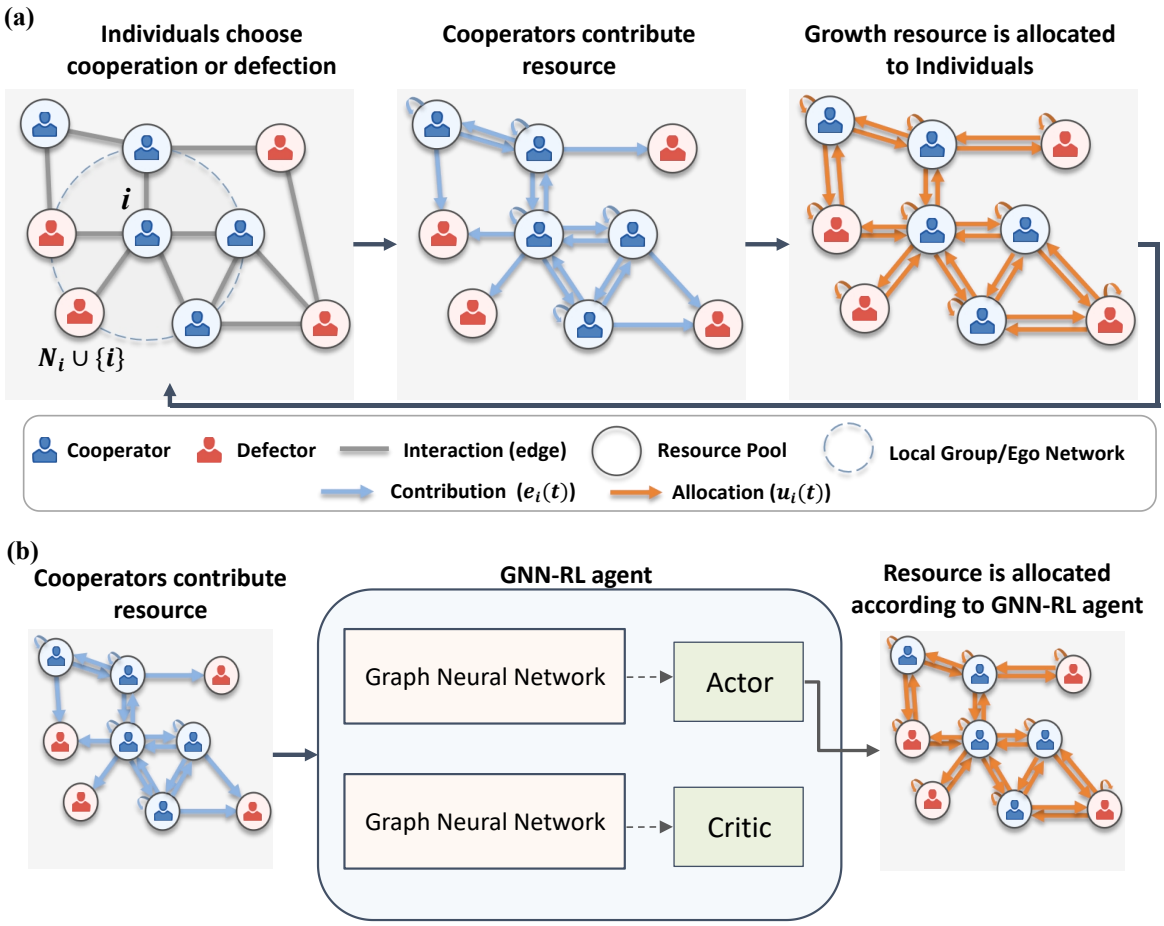

Figure 1: Overview of the network common-pool resource game. (a): In the network common-pool resource game, players are connected through a social network and decide whether to cooperate with or defect against their neighbours. Cooperators will contribute a certain amount of resources to the resource pools; the resources in the pools grow and are returned to players in their local group according to the allocation mechanism. (b): Our GNN-RL agent learns to act as a social planner, determining how to allocate resources for each resource pool in every round. It uses graph neural networks (GNN) to model the network common-pool game and employs reinforcement learning to optimize the allocation mechanism.

Under the equal baseline, the allocable resources in each pool are uniformly distributed among all individuals in the corresponding ego networks. Specifically, individual i receives $\frac{G_j(t)}{d_j+1}$ resource from each resource pool $j \in N_i \cup \{i\}$ of its ego network. Therefore, the total resources received by individual i from all the pools are given by

$$u_i^{\text{Equal}}(t) = \sum_{j \in N_i \cup \{i\}} \frac{G_j(t)}{d_j + 1}. \tag{4}$$

Under the proportional baseline, allocable resources in each pool are distributed according to the individuals' contributions within that ego network. Specifically, individual i receives $\frac{e_j(t)/(d_j+1)}{\sum_{k \in N_i \cup \{i\}} e_k(t)/(d_k+1)} G_i(t)$ resource from each resource pool $j \in N_i \cup \{i\}$ of its ego network. Accordingly, the total resources received by individual i are

$$u_i^{\text{Prop}}(t) = \sum_{j \in N_i \cup \{i\}} \frac{e_i(t)/(d_j + 1)}{\sum_{k \in N_j \cup \{j\}} e_k(t)/(d_k + 1)} G_j(t). \tag{5}$$

2.3. Resource update and strategy evolution

As mentioned before, individual i contributes $e_i(t)$ to and receives resource $u_i(t)$ from a total of $d_i + 1$ resource pools in its ego network. In addition to resource inputs and outputs, each individual is subject to endogenous resource

consumption, which captures the maintenance cost of sustaining local interactions and holding accumulated resources. It also motivates individuals to acquire resources continuously; otherwise, all individuals will maintain defection when they have sufficient resources that will neither decrease nor be consumed. Specifically, individual i incurs a baseline consumption level proportional to its local interaction scale, together with an additional proportional consumption on the resource amount exceeding this baseline, d_i. The consumption function is defined as

$$c_i(t) = d_i + \tau \max\{0,\ R_i(t) - d_i\}, \tag{6}$$

where $\tau \in [0, 1]$ is the proportional consumption rate for the surplus resource beyond the baseline level d_i. Accordingly, the resource level of individual i updates according to

$$R_i(t+1) = R_i(t) + u_i(t) - e_i(t) - c_i(t). \tag{7}$$

During the evolutionary process, the minimum value of $R_i(t)$ is set to 0; if $R_i(t)$ is less than $d_i + 1$, the individual cannot choose the cooperation strategy either.

Furthermore, the payoff of an individual i is defined as the difference between income and input:

$$r_i(t) = u_i(t) - e_i(t). \tag{8}$$

The strategies evolve dynamically on the Fermi update rule according to the individuals' payoff. At each time step t, individual i randomly selects a neighbour $j \in N_i$ and adopts its strategy with probability

$$P(x_i \leftarrow x_j) = \frac{1}{1 + \exp\left((r_i(t) - r_j(t))/\kappa\right)}, \tag{9}$$

where $\kappa > 0$ controls the level of stochasticity. We illustrate the common-pool resource game in Fig. 1(a).

Table 1: Parameters used in the network common-pool resource game.

Parameter	Description	Value(s)
T	Simulation time steps	2×10^4
N	Network size	50
$\langle k \rangle$	Average node degree	4
p_{ER}	Edge probability of ER networks	$8.16 \times 10^{-}2$
m	Attachment parameter of BA networks	2
k_{WS}	Initial degree of WS networks	4
q_{WS}	Rewiring probability of WS networks	0.1
R_0	Initial resource of each individual	20
r	Pool growth factor	0.5
p_c	Pool-capacity coefficient	50
α	Resource contribution ratio	0.5
τ	Resource consumption rate	0.05
κ	Fermi stochasticity parameter	1

2.4. Evaluation Metrics

To evaluate the system's performance from multiple perspectives, we consider three key metrics: the cooperation level, the average accumulated resources, and the resource inequality. First, the cooperation level is defined as the fraction of cooperators in the population at time t, denoted by $f_c(t)$, which reflects the overall level of cooperative behaviour in the game system. Second, we measure the efficiency of the system by the average accumulated resources:

$$\bar{R}(t) = \frac{1}{N} \sum_{i=1}^{N} R_i(t).$$

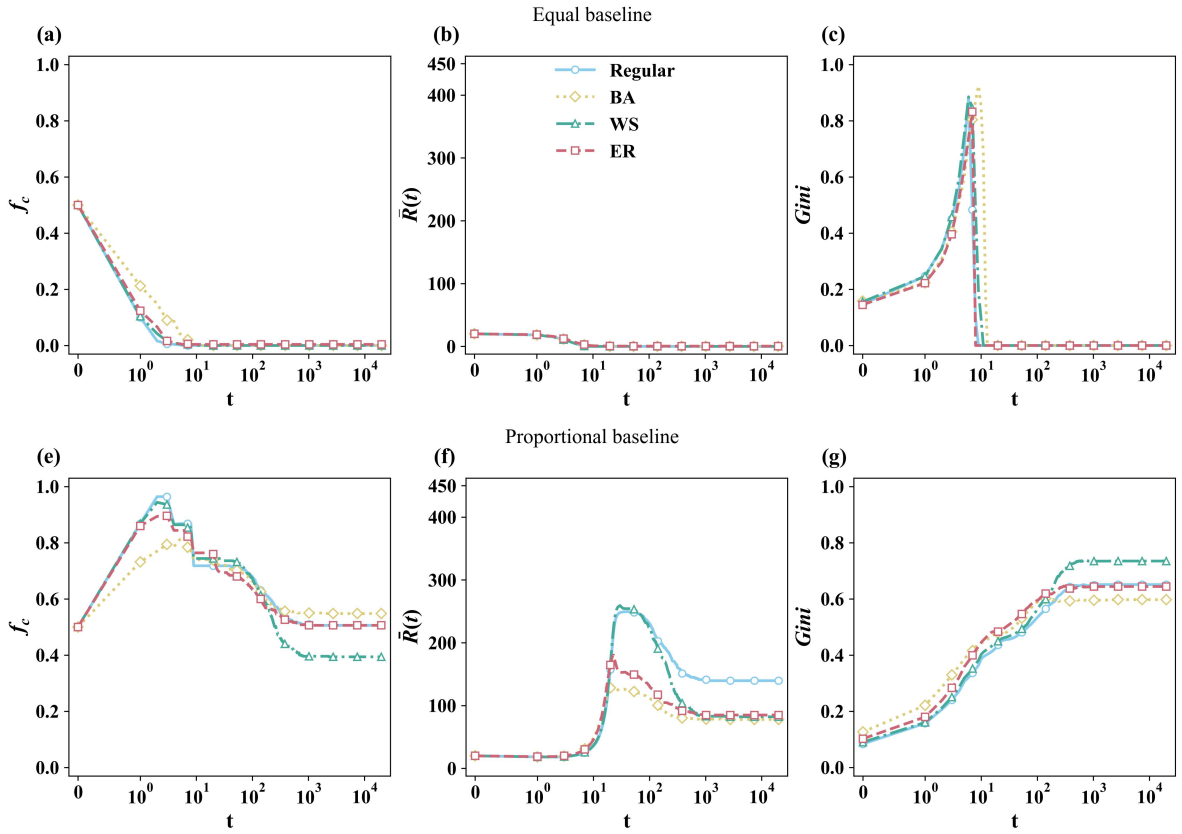

Figure 2: Evaluation metrics over time under different network topologies. (a)-(c) are under the equal baseline. (d)-(f) are under the proportional baseline. Curves are averaged over multiple independent evaluation runs. The equal baseline undermines initiative and encourages free-riding, thereby weakening cooperation; the proportional baseline, on the other hand, can promote cooperation to a certain extent, but subsequently leads to the rich getting richer and the poor getting poorer due to the Matthew effect, thereby reducing the overall level of cooperation.

This metric captures the overall productivity and sustainability of the game system. Finally, we quantify the fairness of resource distribution using the Gini coefficient, defined as

$$Gini = \frac{1}{2N^2\bar{R}(t)} \sum_{i=1}^{N} \sum_{j=1}^{N} |R_i(t) - R_j(t)|,$$

which measures the inequality of resource allocation among individuals. A lower value of *Gini* indicates a more equitable distribution. In numerical implementation, the denominator is replaced by $2N^2(\bar{R}(t) + \epsilon)$ with a small $\epsilon > 0$. Together, these three metrics provide a comprehensive evaluation of cooperation, efficiency, and fairness in the game system.

3. From baseline allocation mechanisms to learning social planner

3.1. Baseline mechanisms and preliminary observations

Before introducing the proposed reinforcement learning framework in Fig. 1(b), we first consider two representative baseline allocation mechanisms to provide reference points for evaluating the effectiveness of learned strategies. Specifically, we examine the equal baseline in Fig. 2(a)-(c) and the proportional baseline in Fig. 2(d)-(f). We evaluate both mechanisms under four different network topologies, including regular networks, Erdős–Rényi (ER) random networks, Barabási–Albert (BA) scale-free networks, and Watts–Strogatz (WS) small-world networks. Unless otherwise specified, we set the evolutionary dynamics parameters as shown in Table 1. To ensure a fair comparison across

different network topologies, the average node degree is fixed at $\langle k \rangle = 4$ in all cases. Moreover, initial strategies are assigned randomly with $f_C(0) = f_D(0) = 0.5$. The results are obtained by averaging multiple independent simulations.

Under the equal allocation mechanism, resources are distributed uniformly, regardless of individual contributions. This design produces a highly equitable distribution outcome, as reflected by consistently low Gini coefficients across all topologies. However, uniform allocation eliminates individual incentives for contribution. Individuals always tend to free-ride. Consequently, individuals gradually lose motivation to cooperate, resulting in a rapid decline in cooperation levels in Fig. 2(a). Ultimately, the system collapses into a state dominated by defection, where cooperative behaviour is no longer sustained. As shown in Fig. 2(b)(c), the average accumulated resources $\bar{R}(t)$ have fallen to near 0, meaning everyone is equally poor. In the Regular, WS, and BA networks, the Gini and the f_c approach 0, indicating that almost no one cooperates and everyone is equally poor; in the ER network, the Gini coefficient is close to 1, but f_c and $\bar{R(t)}$ remain close to 0, meaning that only a very small number of individuals possess resources, while everyone else has none. This observation highlights a fundamental limitation of purely fairness-oriented allocation: while equality is achieved, the lack of incentive mechanisms undermines the sustainability of cooperation.

By contrast, the proportional allocation mechanism rewards individuals based on their contributions, thereby providing a direct incentive for cooperation. Initially, this mechanism promotes cooperative behaviour, leading to a rapid increase in the level of cooperation in Fig. 2(e) from $t = 0$ to $t = 10$. However, the Matthew effect gradually becomes apparent over time, resulting in a decline in the level of cooperation within the system. Individuals with more accumulated resources can invest more resources and thereby acquire even more as a result. Even when the resource pool is saturated, they are less likely to fall into the poverty trap—that is, to lack sufficient resources for cooperation—precisely because they possess more resources. A small number of highly contributing individuals accumulate a disproportionate share of resources, resulting in increased inequality, as indicated by the steadily rising Gini coefficient in Fig. 2(g). Furthermore, despite maintaining a certain level of cooperation, the system's overall performance remains suboptimal. The average accumulated resource $\bar{R}(t)$ first increases and then decreases in Fig. 2(f), and does not reach its theoretical benchmark (see Appendix B), indicating inefficiencies in resource utilization. These results suggest that, while contribution-based allocation can sustain cooperation to some extent, it can also lead to unequal resource distribution and inefficient resource utilization. Since neither the equal baseline nor the proportional baseline can maintain a high level of cooperation while ensuring a relatively fair allocation of resources, we employ a reinforcement learning approach to learn a new allocation mechanism as shown in Fig. 1(b).

3.2. Reinforcement learning framework

To learn an adaptive resource allocation mechanism, we introduce a graph neural network-based reinforcement learning (GNN-RL) framework, as illustrated in Fig. 3. Different from traditional evolutionary game models in which the allocation mechanism is manually specified, the proposed framework acts as a social planner, learning an allocation mechanism from repeated interactions with the network common-pool resource environment. The GNN-RL agent does not directly determine whether individuals cooperate or defect. Instead, it acts as a social planner. At each time step, every individual first chooses their strategies according to the evolutionary game process described in Section 2. Cooperators contribute resources to local pools, and the GNN-RL agent then determines how the grown pool resources should be allocated back to the individuals in each ego network, as shown in Fig. 1(b). The framework consists of three main modules: an observation graph module, an actor network, and a critic network.

At each time step, the current game system state is represented as the environment state, and we represent the current game state as a graph tensor representation (u, V, E). The global feature vector $u \in \mathbb{R}^4$ contains graph-level summary statistics, including the fraction of cooperators f_c, the mean accumulated resource $\bar{R}(t)$, the mean resource pool value $\bar{G}(t)$, and the Gini coefficient. The node feature matrix $V \in \mathbb{R}^{N\times 4}$ contains, for each node i, its resource pool $G_i(t)$, accumulated resource $R_i(t)$, degree d_i, and the contribution resource $e_i(t)$. The edge feature tensor $E \in \mathbb{R}^{N\times N\times 1}$ records the connectivity information used by the model. Specifically, each entry indicates whether node j is included in the feasible ego-network of node i.

The actor network is responsible for generating allocation actions. It first encodes the global structural and state information of the network using a two-layer GraphNet backbone. Then, for each focal node i, which defines a local resource pool over its ego-network, a shared ego-network allocation head is applied to the subgraph consisting of node i and its neighbours. Based on the learned local representations, the actor produces allocation scores for all feasible participants in the pool, and a softmax operation normalizes these scores into allocation weights. Repeating this process for all focal nodes yields the final allocation action over all local pools. This design matches the local interaction

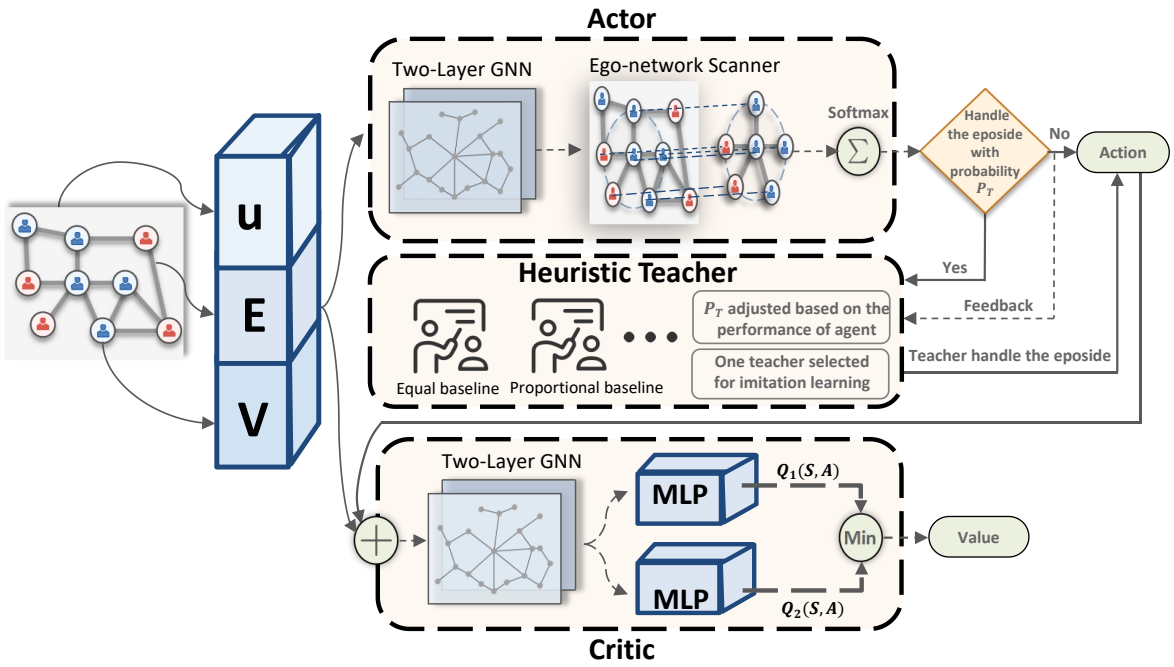

Figure 3: GNN-RL Agent architecture. The agent observes global features u, node features V, and edge features E from the game state. The actor first encodes the graph with a two-layer GraphNet backbone, then applies a shared ego-network allocation head to each focal node, functioning like a convolution kernel. Due to the randomness of evolutionary games and their sensitivity to allocation strategies, we introduce heuristic teachers' strategies to accelerate and stabilize the learning process.

structure of the common-pool resource game. Since each resource pool is defined within a local neighbourhood, the ego-network allocation head enables pool-level decision making while sharing parameters across the entire graph.

The critic network evaluates the quality of the allocation actions generated by the actor. It takes the current graph state and the allocation action as input, and estimates the expected long-term return. To improve training stability and reduce overestimation bias, we adopt a twin-critic structure. Two critic networks independently estimate the action value, and the smaller value is used during policy evaluation. This design follows the common practice in actor–critic algorithms for continuous control problems.

3.3. Training design

Training the social planner in the proposed evolutionary game is inherently challenging because the environment is highly non-stationary. Individuals continuously update their strategies during an episode, and the planner's allocation decisions influence not only the immediate state but also the subsequent evolutionary trajectory of the system. In particular, poor allocation decisions in the early stage of an episode may drive the population into a low-cooperation regime that is difficult to recover from later. To limit this long-horizon instability and improve optimization tractability, we use relatively short episodes of 200 steps. This choice is also supported by preliminary simulations, in which representative heuristic baselines such as equal and proportional allocation already exhibit stable convergence behaviour within this horizon.

To improve training stability and robustness, we train the planner over multiple graph topologies rather than a single fixed interaction structure. Specifically, the training distribution comprises four topology families, including regular networks, ER networks, BA networks, and WS networks. For each topology, we construct a fixed graph bank using predetermined random seeds. During training, graph instances are sampled from these fixed banks. This design increases structural diversity while avoiding the excessive variance that would arise if entirely new graphs were regenerated at every reset. As a result, the planner is exposed to heterogeneous network structures while the optimization process remains reproducible and more stable.

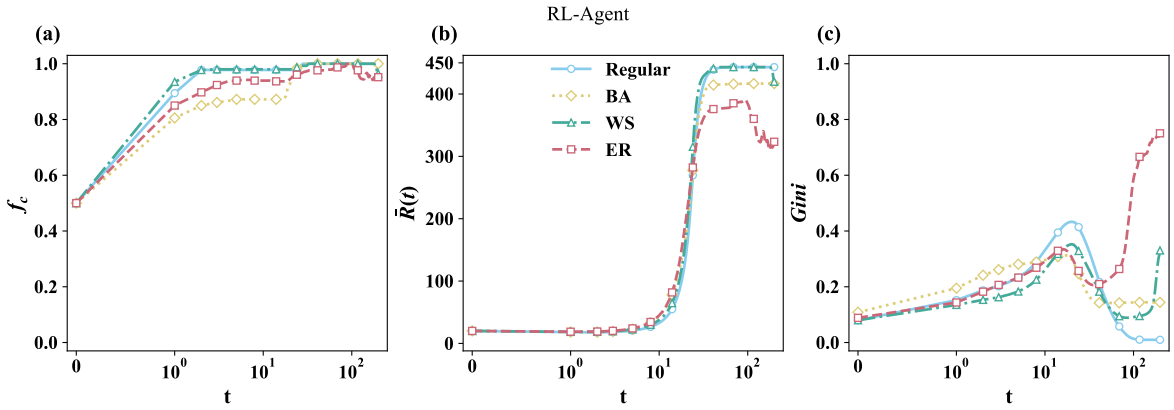

Figure 4: Evaluation metrics over time under different network topologies for the RL-agent. Here, we used the same topology seeds and the episode length ($T = 200$) as during training. Curves are averaged over multiple independent evaluation runs. The strategy of the RL-Agent can learn a good allocation strategy to significantly enhance the level of cooperation, accumulated resources, and reduce the Gini index, despite some fluctuations at the end of the round on certain types of networks.

To further reduce the instability caused by uninformative exploration, we adopt a staged teacher-guided training procedure. The teachers consist of equal baseline, proportional baseline and a pool-dependent heuristic baseline that adaptively combines uniform and contribution-proportional allocation according to the current pool state [49]. During online TD3 training, teacher-guided actions are initially retained to prevent early cooperation collapse. Teacher control is then released gradually after the actor demonstrates sufficiently stable performance. At the same time, the imitation-learning constraint is progressively weakened and the actor's value-maximization objective is strengthened. This staged handoff allows the planner to benefit from a stable initial allocation mechanism while eventually learning an autonomous policy that can improve beyond the heuristic teacher. The actor architecture is further designed to match the local interaction pattern of the common-pool resource game. After global message passing over the entire graph, the final policy head is implemented as a shared ego-network allocation head, which is applied repeatedly to each focal node and its local neighbourhood. In this way, each focal node defines a local resource pool, and the actor produces a row-wise allocation distribution over the corresponding local participant set. This convolution-inspired design introduces a strong locality prior: allocation decisions are made at the pool level, while the same local decision mechanism is shared across all nodes in the graph. Besides improving generalization across graph instances, this design also makes the learned policy easier to interpret, since the final action can be understood as repeated applications of a common local allocation mechanism. In other words, a social planner applies the same allocation mechanism to the ego network of every node in the entire graph. For the optimization objective, we use a resource-oriented planner reward. At each step, the planner receives a scaled reward proportional to the population-mean resource level in the next state, denoted by $\lambda_R \bar{R}(t+1)$. The training objective is to maximize the cumulative return over the episode. This reward design is motivated by the observation that persistently high resource levels can only be maintained when the population remains in a highly cooperative regime. Therefore, although the reward is defined in terms of system-wide resources, it also serves as an indirect objective for sustaining cooperation over time. We adopt the standard TD3 update scheme, including delayed policy updates and soft target-network updates. The detailed network architecture and training hyperparameters are provided in Appendix A.

Overall, the proposed GNN-RL framework learns an adaptive allocation mechanism over structured populations. By combining graph neural networks with reinforcement learning, the model can exploit local topological information while optimizing long-term collective outcomes.

4. Performance and interpretation of the RL agent's policy

4.1. Performance of the learned GNN-RL social planner

We first evaluate the performance of the learned GNN-RL social planner in the training environment. The episode length is 200, and the topology seeds are the same as those used during training. Fig. 4 shows the temporal evolution

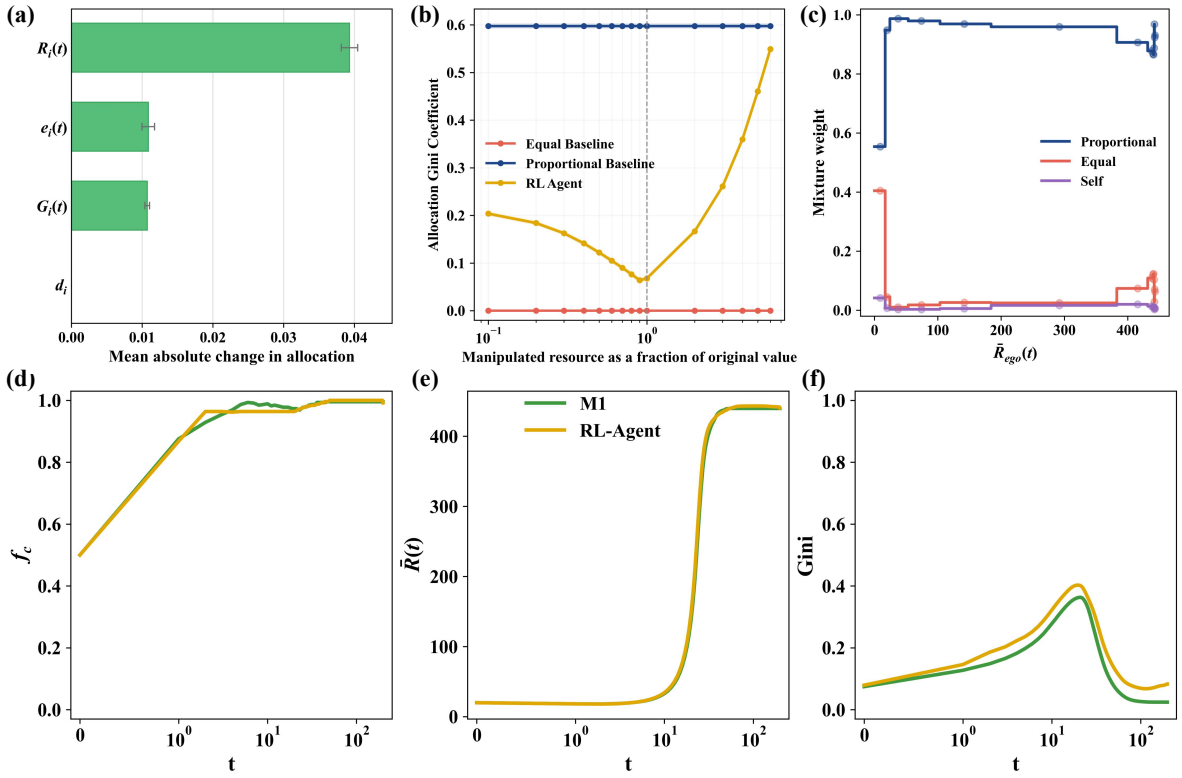

Figure 5: Interpretation of the learned allocation mechanism on regular networks. (a) Counterfactual feature-importance analysis of the actor's allocation output. Error bars denote SEM (Standard Error of the Mean) over 1,000 state snapshots. (b) Single-variable intervention on accumulated resource, where the resource feature is manipulated as a fraction of its original value. (c) Fitted mixture weights of the resource-binned mixture mechanism as a function of the ego-network average accumulated resource $\bar{R}_{\text{ego}}(t)$. Solid step lines denote the fitted mixture weights, and circular markers denote the corresponding empirical estimates/data points. (d)–(f) Comparison between the original RL-Agent and the interpretable mixture mechanism M1 in terms of cooperation level, average accumulated resource, and Gini coefficient. Here, we used the same topology seeds and the length of episode ($T = 200$) as during training. Results are averaged over multiple independent evaluation runs.

of the cooperation level $f_c(t)$, the average accumulated resource $\bar{R}(t)$, and the Gini coefficient under different topologies. In each panel, different curves correspond to different network topologies, including Regular, ER, BA, and WS networks. For the WS and ER networks, $\bar{R}(t)$ and the Gini coefficient exhibit visible fluctuations during the later stage of the episode, especially for $t = 100$–200. This is because the curves are averaged over multiple independent evaluation runs, and a small number of runs under these two topologies show relatively poor performance in the later stage. Nevertheless, compared with the equal baseline, the GNN-RL planner maintains a much higher cooperation level and prevents the system from rapidly collapsing into a low-resource state. This suggests that the learned planner preserves stronger incentives for cooperation than purely uniform redistribution. Compared with the proportional baseline, the GNN-RL planner achieves a more balanced outcome. It sustains a higher level of accumulated resources while maintaining a lower Gini coefficient.

Overall, the learned GNN-RL social planner improves cooperation, resource sustainability, and fairness across different network topologies. These results indicate that the trained planner does not simply reproduce either equal or proportional allocation. Instead, it learns an adaptive allocation policy that can balance cooperation incentives and resource redistribution according to the evolving system state.

4.2. Interpretation of the learned allocation mechanism

After evaluating the performance of the learned GNN-RL social planner, we further investigate what allocation mechanism it has learned. We first focus on regular networks, where all nodes have the same degree. This setting removes the influence of degree heterogeneity and allows us to identify the state variables that directly affect the

learned allocation policy.

We begin with a counterfactual feature-importance analysis. For each input feature in the ego-network representation, we set its value to zero while keeping the remaining features unchanged, and then measure the resulting change in the actor's allocation output. As shown in Fig. 5(a), the accumulated resource $R_i(t)$ has the strongest influence on the learned allocation decisions. In contrast, the contribution resource $e_i(t)$, the grown pool value $G_i(t)$, and the degree d_i have weaker effects. The weak effect of degree is expected in regular networks, since all nodes have the same structural role. To further verify the role of accumulated resource, we conduct a single-variable intervention by manipulating the accumulated resource feature as a fraction of its original value. In this intervention, only the resource feature observed by the actor is modified. The actual contribution $e_i(t)$ and the environment dynamics are still computed from the original resource level before intervention. Fig. 5(b) shows that the allocation behaviour of the RL agent changes systematically when the accumulated resource feature is manipulated, while the equal and proportional baselines remain fixed because they do not depend on this state variable. The allocation Gini coefficient changes markedly as the accumulated resource feature is manipulated, indicating that the learned actor adjusts the inequality of its allocation according to the perceived local resource state. This result confirms that the learned planner actively uses accumulated resource information when making allocation decisions in the ego networks.

Based on the above observation, we construct an interpretable resource-binned mixture mechanism, denoted by M1. Since the actor applies a shared ego-network allocation head to scanning every focal node, we interpret the learned policy at the ego-network level. For the resource pool centered at node i, we define the mean accumulated resource of its ego-network as

$$\bar{R}_{\mathrm{ego},i}(t) = \frac{1}{d_i + 1} \sum_{j \in N_i \cup \{i\}} R_j(t).$$

This quantity summarizes the local resource state of all individuals participating in pool i. We then divide $\bar{R}_{\mathrm{ego},i}(t)$ into several bins and approximate the actor's allocation behaviour within each bin as a mixture of three representative allocation mechanisms: equal allocation, proportional allocation, and self-allocation.

For the resource pool centered at node i, let $\hat{A}_{i\to j}(t)$ denote the allocation proportion assigned to individual $j \in N_i \cup \{i\}$. Therefore, the actual amount of resources allocated from pool i to individual j is $\hat{A}_{i\to j}(t)G_i(t)$. The allocation proportion of M1 is expressed as

$$\hat{A}^{\mathrm{M1}}_{i\to j}(t) = \omega^{(b_i(t))}_{\mathrm{Equal}} \frac{1}{d_i + 1} + \omega^{(b_i(t))}_{\mathrm{Prop}} \frac{e_j(t)/(d_j + 1)}{\sum_{k \in N_i \cup \{i\}} e_k(t)/(d_k + 1)} + \omega^{(b_i(t))}_{\mathrm{Self}} \mathbf{1}_{j=i},$$

where $b_i(t)$ denotes the bin index determined by $\bar{R}_{\mathrm{ego},i}(t)$. For each bin, the three mixture weights are constrained to be non-negative and sum to one. The three coefficients $\omega^{(b_i(t))}_{\mathrm{Equal}}$, $\omega^{(b_i(t))}_{\mathrm{Prop}}$, and $\omega^{(b_i(t))}_{\mathrm{Self}}$ represent the relative weights of equal allocation, proportional allocation, and self-allocation in the corresponding resource bin, respectively.

Fig. 5(c) shows the fitted mixture weights under different ego-network resource bins defined by $\bar{R}_{\mathrm{ego},i}(t)$. The learned policy is not equivalent to a purely equal or purely proportional mechanism. Instead, we interpret it as a resource-dependent mixture of several simple allocation principles. In regular networks, the proportional component remains dominant over most ego-network resource bins, while the equal and self-allocation components provide additional state-dependent adjustments. The mixture weight of equal allocation increases in both low- and high-resource regimes, suggesting that the RL agent introduces stronger redistribution when the local resource state is either fragile or sufficiently abundant.

Finally, we evaluate whether this interpretable mixture mechanism can reproduce the performance of the original RL agent. As shown in Fig. 5(d)–(f), M1 produces dynamics close to those of the RL-Agent in terms of cooperation level, average accumulated resource, and Gini coefficient. Although M1 is much simpler than the neural policy, it preserves the main performance advantage of the learned planner. This suggests that, on regular networks, the main behaviour of the GNN-RL planner can be captured by a resource-dependent mixture mechanism.

We next examine how the learned allocation mechanism behaves when structural heterogeneity is introduced. For this purpose, we focus on scale-free networks, where node degrees are highly heterogeneous, and a small number of hub nodes coexist with many low-degree nodes. This setting allows us to investigate whether the learned planner treats individuals differently according to their structural positions.

To quantify the degree-dependent allocation pattern, we divide each trajectory into three temporal stages: the first 20%, the middle 60%, and the last 20% of the episode. Within each stage, we compute the accumulated incoming

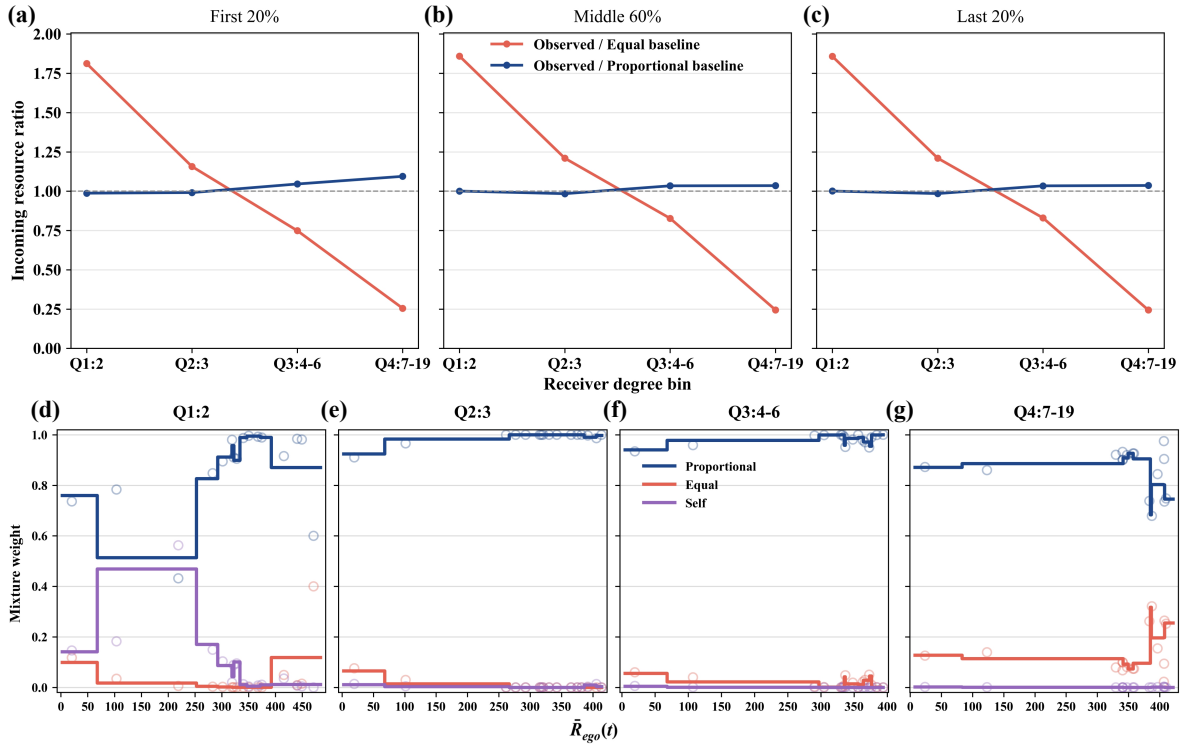

Figure 6: Interpretation of the learned allocation mechanism on scale-free networks. (a)-(c): degree-dependent incoming resource ratios in scale-free networks. Nodes are divided into four bins based on their degree: Q1 consists of nodes with a degree of 2, Q2 consists of nodes with a degree of 3, Q3 consists of nodes with a degree between 4 and 6, and Q4 consists of nodes with a degree between 7 and 19. Each trajectory is divided into three temporal stages: the first 20%, the middle 60%, and the last 20% of the episode. For each stage, nodes are grouped by receiver degree into four bins. The red curve shows the ratio of observed incoming resources under the learned RL agent to those under the equal baseline. The blue curve shows the ratio of observed incoming resources under the learned RL agent to those under the proportional baseline. A ratio larger than 1 indicates that the corresponding degree bin receives more resources than the baseline, whereas a ratio smaller than 1 indicates that it receives less. (d)-(g): degree-conditioned mixture mechanism M2 in scale-free networks. The four panels correspond to different focal-node degree bins: Q1:2, Q2:3, Q3:4–6, and Q4:7–19. For each degree bin, the horizontal axis represents the ego-network average accumulated resource $\bar{R}_{\mathrm{ego}}(t)$, and the vertical axis represents the mixture weight of each allocation component. The three components are proportional allocation, equal allocation, and self-allocation. Circular markers denote binned empirical weights estimated from the RL-agent rollout, while solid step curves denote the fitted discrete mechanism used by M2.

resources received by each node. Nodes are then grouped according to their receiver degree into four bins: Q1:2, Q2:3, Q3:4–6, and Q4:7–19. For each bin, we report how much resource the learned RL-agent allocates relative to the Equal and Proportional baselines. A ratio larger than 1 indicates that the corresponding degree bin receives more resources than the baseline prediction, whereas a ratio smaller than 1 indicates that it receives less. As shown in Fig. 6(a)-(c), the observed allocation differs strongly from the equal baseline. Low-degree nodes receive substantially more resources than they would under equal allocation, while high-degree nodes receive much less. This suggests that the learned planner does not simply amplify the structural advantage of high-degree nodes, even though such nodes participate in more local pools. By contrast, the observed allocation is much closer to the proportional baseline. The ratio relative to proportional allocation remains close to 1 across degree bins and temporal stages, with only mild deviations for higher-degree nodes. This indicates that, in heterogeneous degree networks, the learned policy preserves a contribution-related allocation structure while adjusting it according to degree heterogeneity. Importantly, this degree-dependent pattern is relatively stable across the first, middle, and last stages of the episode. Therefore, the degree effect should be interpreted mainly as a structural feature of the learned allocation mechanism, rather than as a transient temporal effect. This motivates the following degree-binned analysis, where we explicitly incorporate receiver degree into the interpretable mixture mechanism.

Based on the above degree-dependent analysis, we further construct a degree-conditioned mixture mechanism for

heterogeneous networks, denoted by M2. Compared with M1, where the mixture weights depend only on the ego-network average accumulated resource, M2 additionally conditions the mixture weights on the degree of the focal node. This extension is motivated by the observation that, in scale-free networks, nodes with different degrees play different structural roles and may require different local allocation mechanisms.

For each resource pool centered at node i, we use the ego-network average accumulated resource $\bar{R}_{\mathrm{ego},i}(t)$ to characterize the local resource state, and use the degree d_i of the focal node to characterize its structural role. The focal nodes are divided into four degree bins, denoted by Q1:2, Q2:3, Q3:4–6, and Q4:7–19. Within each degree bin, we fit a resource-dependent mixture mechanism over three representative allocation components: proportional allocation, equal allocation, and self-allocation. Formally, the M2 allocation proportion from pool i to individual $j \in N_i \cup \{i\}$ is written as

$$\hat{A}^{\mathrm{M2}}_{i\to j}(t) = \omega^{(q_i,b_i(t))}_{\mathrm{Prop}} \frac{e_j(t)/(d_j+1)}{\sum_{k\in N_i\cup\{i\}} e_k(t)/(d_k+1)} + \omega^{(q_i,b_i(t))}_{\mathrm{Equal}} \frac{1}{d_i+1} + \omega^{(q_i,b_i(t))}_{\mathrm{Self}} \mathbf{1}_{j=i},$$

where q_i denotes the degree bin of the focal node i, and $b_i(t)$ denotes the resource bin determined by $\bar{R}_{\mathrm{ego},i}(t)$. For each pair of bins $(q_i, b_i(t))$, the three mixture weights are non-negative and sum to one. Fig. 6(d)-(h) visualizes the fitted degree-conditioned mixture mechanism. The four panels correspond to different focal-node degree bins. Within each panel, the horizontal axis represents the ego-network average accumulated resource $\bar{R}_{\mathrm{ego}}(t)$, and the vertical axis represents the fitted mixture weight of each allocation component. The circular markers show the binned empirical weights estimated from the RL-agent rollout, while the solid step curves show the discrete mechanism used by M2.

The fitted profiles reveal that the learned allocation mechanism changes substantially across degree groups. For low-degree focal nodes, the self-allocation component can become important in some intermediate resource regimes. This indicates that the learned planner tends to preserve resources around structurally peripheral nodes. In contrast, for middle-degree nodes, the proportional component dominates almost all resource regimes, suggesting that contribution-based allocation is sufficient for these nodes. However, when total resources are low, a certain equal baseline coefficient is applied to prevent individuals from falling into the poverty trap. For high-degree focal nodes, the equal component becomes more pronounced when the ego-network resource level is high, indicating that the planner introduces stronger redistribution around hub-like local pools. These results suggest that, in heterogeneous networks, the learned policy cannot be fully captured by a resource-only mechanism. Instead, it is better interpreted as a degree-conditioned mixture mechanism that adapts allocation principles according to both local resource states and structural roles. Subsequent simulations further demonstrated this point.

To further evaluate the performance of the interpretable mixture mechanisms M1 and M2, we tested five allocation mechanisms on newly generated random graph instances in Fig. 7. Unlike the evaluation conducted during training, the graph instances used here are fully resampled and are not taken from the fixed topology bank used during RL training. This setting provides a stricter test of generalization to unseen network structures. In addition to the randomly sampled network evaluation reported here, we further test the five mechanisms under parameter shifts in p_c, κ, and r (see Appendix C). Fig. 7 compares the equal baseline, the proportional baseline, the RL-Agent, the resource-binned mixture mechanism M1, and the degree-conditioned mixture mechanism M2 across four network topologies. The three columns report the cooperation level $f_c(t)$, the average accumulated resource $\bar{R}(t)$, and the Gini coefficient, respectively. The Equal baseline rapidly collapses in all network topologies, confirming that purely uniform redistribution cannot maintain cooperation. The Proportional baseline performs better than the Equal baseline at the early stage, but it eventually leads to declining cooperation, reduced accumulated resources, and high inequality, especially in heterogeneous networks. The resource-binned mechanism M1 performs well on regular networks, where degree heterogeneity is absent. However, its performance deteriorates on BA, ER, and WS networks. This result is consistent with the previous interpretation: M1 only conditions the allocation mechanism on the ego-network resource level and therefore cannot fully account for structural heterogeneity. The original RL-Agent also exhibits strong performance on several topologies, but its dynamics become less stable on unseen random graphs. In particular, under ER and WS networks, the RL-Agent initially improves cooperation and accumulated resources, but the performance later declines, accompanied by a marked increase in inequality. This suggests that the RL-Agent is sensitive to distributional changes in graph structure and to the long-term stochastic evolutionary process when evaluated on more diverse network instances and over longer evolutionary horizons than those used during training.

By contrast, M2 achieves the most robust overall performance across all four topologies. It maintains a high cooperation level and high accumulated resources in regular, BA, ER, and WS networks, while keeping the Gini

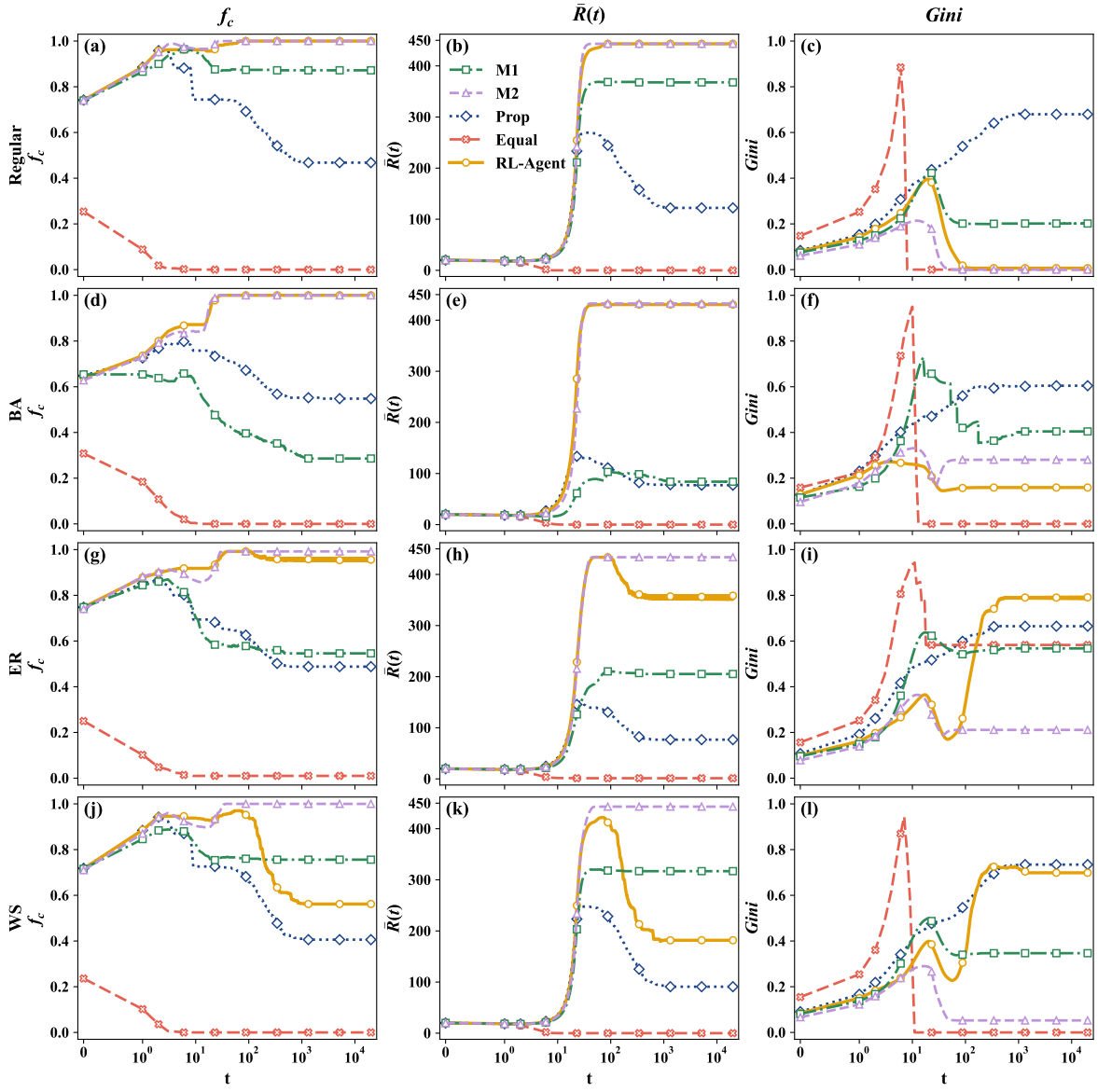

Figure 7: Evaluation metrics over time in four topologies under different allocation mechanisms. Rows correspond to (a)-(c): Regular, (d)-(f): BA, (g)-(i): ER, and (j)-(l): WS networks. Columns correspond to f_c, $\bar{R}(t)$, and *Gini*, respectively. The compared mechanisms are M1, M2, Proportional (Prop), Equal, and the RL-Agent. The horizontal axis shows time t on a logarithmic scale ($T = 2 \times 10^4$). The results are obtained by averaging 10 independent simulations.

coefficient relatively low. This indicates that explicitly incorporating the degree-conditioned allocation mechanism improves the generalization of the learned mechanism. In other words, M2 captures not only the resource-dependent pattern identified in regular networks, but also the structural adjustment required in degree-heterogeneous networks.

To further interpret the results, we compare the observed Gini coefficients with the full-cooperation equal-allocation benchmarks derived in Appendix B. These benchmarks describe the level of resource inequality induced purely by degree heterogeneity under full cooperation and equal local redistribution. Under the default parameter setting, the corresponding Gini benchmarks are approximately 0.3117 for BA networks, 0.2122 for ER networks, and 0.0564 for WS networks. Interestingly, M2 achieves Gini coefficients of approximately 0.2805, 0.2111, and 0.0525 on BA, ER, and WS networks, respectively, while maintaining high cooperation levels and near-optimal average accumulated

resources. In particular, the improvement is most evident in the BA scale-free network, where degree heterogeneity is strongest. This result suggests that equal allocation does not necessarily imply global fairness in heterogeneous networks. Even if every local pool distributes resources equally, nodes with different degrees participate in different numbers of pools, which can generate systematic inequality at the population level. This also indicates that purely equal allocation is not necessarily globally fair in degree-heterogeneous networks. Instead, a structure-aware allocation mechanism can compensate for degree-induced inequality by adapting redistribution to network position.

The degree-conditioned structure of M2 provides an interpretable explanation for this improvement. For low-degree focal nodes, M2 assigns a higher weight to the self-allocation component in some resource regimes, which helps peripheral nodes preserve their own resources and maintain the ability to continue cooperating. For middle-degree nodes, the proportional component dominates, indicating that contribution-based incentives are sufficient to sustain cooperative development. At the same time, M2 maintain a certain degree of equal baseline allocation when ego-network resources are scarce to prevent individuals from falling into the poverty trap. For high-degree nodes, the equal-allocation component becomes more pronounced, suggesting that hub-like nodes should place more emphasis on supporting their surrounding neighbours rather than only reinforcing their own contribution advantage. Therefore, M2 can be interpreted as a structure-aware allocation mechanism: peripheral nodes are protected from falling into poverty traps, middle-degree nodes are incentivized through proportional rewards, and high-degree nodes help redistribute resources within their larger local neighbourhoods.

5. Conclusions

This study investigated how adaptive allocation mechanisms can promote cooperation in resource-constrained networked populations. We proposed a network common-pool resource game in which individuals are embedded in complex networks, participate in overlapping local resource pools, and face endogenous resource constraints during strategy evolution. This framework captures the coupling among cooperation, resource accumulation, local interaction structure, and allocation rules.

Our results show that simple hand-designed allocation mechanisms face inherent limitations. Equal allocation produces fair but inefficient outcomes, because it weakens contribution incentives and may lead to cooperation collapse. Proportional allocation can temporarily stimulate cooperation, but it amplifies accumulated advantages and may push resource-poor individuals into poverty traps. By contrast, the proposed GNN-RL social planner learns adaptive allocation policies that achieve more balanced outcomes across different network topologies, sustaining higher cooperation levels, maintaining larger average accumulated resources, and avoiding the severe inequality induced by purely proportional allocation. More importantly, the learned policy reveals interpretable principles for resource sharing. The extracted mixture mechanisms show that effective allocation should not be fixed, but should depend on both the local resource state and the structural role of the resource pool. On regular networks, the learned policy can be approximated by a resource-dependent mixture mechanism. On heterogeneous networks, the degree-conditioned mixture mechanism further shows that peripheral nodes, intermediate-degree nodes, and hub-like nodes require different combinations of protection, contribution-based reward, and redistribution. These findings suggest that sustainable cooperation can benefit from allocation mechanisms that are sensitive to both local scarcity and network position. This point is further supported by the steady-state analysis in the Appendix, which shows that even under persistent global cooperation and equal local redistribution, heterogeneous networks can still induce nonzero resource inequality through degree heterogeneity.

These results provide implications for public goods provision, ecological resource management, platform governance, and organizational incentive design. A mechanism that appears fair in the short term may fail if it removes incentives to contribute, while a mechanism that strongly rewards contribution may become unsustainable if it allows early advantages to accumulate without correction. In this sense, the learned and extracted mechanisms offer a computational perspective on how artificial intelligence can assist the design of adaptive and interpretable governance rules. Future research may further examine the theoretical stability of the extracted mechanisms, extend the framework to larger and dynamically changing networks, and incorporate richer behavioural assumptions. In particular, since individual strategy updates in the present model are governed by a fixed Fermi rule, future work could introduce human participants or human-behaviour models to evaluate whether the learned allocation mechanisms remain effective under more realistic decision-making processes [48, 49, 51]. More broadly, when AI systems are deployed as

social planners in the real world, their intervention principles should be evaluated not only by aggregate performance, but also by transparency, interpretability, and alignment with human welfare [52–54].

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article

Appendix A. Implementation details of the GNN-RL planner

This appendix provides additional implementation details of the proposed GNN-RL planner. The planner is designed to act as an external social planner that observes the current networked resource state and determines how the allocable resources of each local pool should be distributed. Its implementation consists of two main parts: a graph-based actor network that outputs local allocation weights and twin graph-based critic networks that estimate the action value of the allocation policy.

The detailed network architecture is summarized in Table A.1. The actor first encodes the global network state through stacked GraphNet blocks and then applies a shared ego-network allocation head to each focal node. This design allows the same local allocation rule to be repeatedly applied to different resource pools while still conditioning the decision on the local neighborhood structure and node-level resource states. The critic networks use a similar graph-based state representation but are independently parameterized from the actor, so that policy evaluation and policy improvement remain separated during training.

The training hyperparameters are reported in Table A.2. We use the TD3-style training scheme with twin critics, delayed actor updates, and soft target-network updates. These components are used to reduce overestimation bias and improve training stability in the non-stationary evolutionary environment. During training, each episode contains 200 evolutionary steps, while the evaluation experiments in the main text use longer simulation horizons to examine whether the learned allocation mechanism can sustain cooperation over extended dynamics. The social planner's reward at each step is defined as the next-step average accumulated resource. At each training step, after the planner determines the allocation action and the environment evolves to the next state, the planner receives a scaled next-step average-resource reward:

$$r_{\text{planner}}(t) = \lambda_R \bar{R}(t+1). \tag{A.1}$$

Here, $\lambda_R > 0$ is a constant reward-scaling coefficient used to adjust the numerical magnitude of the reward during training. It should be noted that the reward does not explicitly include the cooperation level or the Gini coefficient. Therefore, the improvements in cooperation and fairness reported in the main text should be interpreted as emergent consequences of the learned allocation policy rather than as directly optimized objectives. Also, due to the properties of network dynamics, achieving a high level of cumulative resources requires a low Gini coefficient and a high level of cooperation.

Appendix B. Analysis of Steady State

In this section, we provide two analytical benchmarks for the model. We first derive an upper bound for the steady-state resource level under a homogeneous approximation, and then analyze the degree-induced Gini coefficient on BA scale-free networks under full cooperation.

Table A.1: Summary of the network architecture used in the proposed GNN-RL planner.

Component	Specification
Input representation	Graph tensor (u, V, E), where u denotes global features, V denotes node features, and E denotes edge features.
First GraphNet block	MLP-based message passing updates for edge, node, and global representations; hidden dimension 128.
Second GraphNet block	MLP-based message passing updates for edge, node, and global representations; hidden dimension 128.
Ego-network extraction	For each focal node, an ego-network consisting of the node and its one-hop neighbours is extracted as a local resource pool.
Local ego-network head	A shared local GraphNet block is applied to each ego-network; hidden dimension 64.
Allocation score head	An MLP-based score readout with hidden dimension 64, followed by softmax normalization to produce row-wise allocation weights.
Actor output	Row-wise local allocations are assembled into the final allocation action over all local resource pools.
Twin critics	Two graph action-value critics with identical architecture and independent parameters.
Critic state encoder	Two stacked GraphNet blocks with hidden dimension 64.
Critic value head	MLP-based local action encoder, pool-level encoder, and scalar Q-value head, all with hidden dimension 64.
Activation and sharing	GELU is used in MLP modules; the ego-network allocation head is shared across all focal nodes.

Notes: (1) The actor global encoder and the critic state encoder use the same input format but do not share parameters. (2) The ego-network allocation head is applied repeatedly to every focal node with shared weights. (3) The twin critics share the same architecture but are independently parameterized.

Table A.2: Main training hyperparameters of the proposed GNN-RL planner.

Parameter	Description	Value
Training steps	Total number of environment steps	5×10^7
Episode length	Number of steps per training episode	200
Batch size	Mini-batch size for replay-buffer sampling	256
Actor learning rate	Learning rate of the actor network	5×10^{-6}
Critic learning rate	Learning rate of the critic networks	2×10^{-5}
γ	Discount factor	0.99
ρ	Target-network soft update rate	0.005
Policy delay	Frequency of delayed actor updates	8
Gradient steps	Gradient steps per environment update	2
Reward signal	Scaled next-step average-resource reward	$\lambda_R \bar{R}(t+1)$
λ_R	Reward scaling coefficient	0.027

Appendix B.1. Steady-state resource analysis

To obtain an analytical benchmark for the resource level under persistent cooperation, we consider a homogeneous setting in which the interaction network is d-regular and all individuals remain cooperative at steady state. In addition, we assume that resource allocation within each local pool is uniform (Equal baseline). Under this approximation, all individuals are statistically equivalent, and we denote their common resource level at time t by $R(t)$.

Since every node has degree d, each individual participates in exactly $d+1$ local pools, including its own pool and

those centered at its neighbours. Under full cooperation, the contribution of each individual is

$$e(t) = d + 1 + \alpha(R(t) - (d + 1)). \tag{B.1}$$

Because the network is homogeneous, each pool receives the same total contribution,

$$P(t) = \sum_{j \in N_i \cup \{i\}} \frac{e_j(t)}{d_j + 1} = e(t). \tag{B.2}$$

The allocable resource in each pool is therefore

$$G(t) = \min\{(1 + r)e(t),\ p_c(d + 1)\}. \tag{B.3}$$

Under uniform allocation, each pool returns $G(t)/(d + 1)$ to each of the $d + 1$ members in its local group. Since each individual belongs to exactly $d + 1$ pools, the total received resource is

$$u(t) = G(t). \tag{B.4}$$

Moreover, under the all-cooperation steady state we must have $R(t) \geq d+1 > d$, and thus the consumption function lies in the linear regime:

$$c(t) = d + \tau(R(t) - d). \tag{B.5}$$

Hence, the resource update equation becomes

$$R(t + 1) = R(t) + u(t) - e(t) - c(t). \tag{B.6}$$

At steady state, $R(t + 1) = R(t) = R^*$, so

$$e^* + c^* = u^* = G^*. \tag{B.7}$$

Since

$$G^* = \min\{(1 + r)e^*,\ p_c(d + 1)\} \leq p_c(d + 1), \tag{B.8}$$

it follows that

$$e^* + c^* \leq p_c(d + 1). \tag{B.9}$$

Substituting the expressions for e^* and c^* yields

$$d + 1 + \alpha(R^* - (d + 1)) + d + \tau(R^* - d) \leq p_c(d + 1). \tag{B.10}$$

After rearranging, we obtain the upper bound

$$R^* \leq \frac{(\tau - 1)d + (p_c + \alpha - 1)(d + 1)}{\alpha + \tau}. \tag{B.11}$$

Therefore, under homogeneous averaging and persistent cooperation, the steady-state resource level of an individual is bounded above by Eq. (B.11). Equality holds if and only if the local pool operates on the saturated branch,

$$(1 + r)e^* \geq p_c(d + 1), \tag{B.12}$$

so that

$$G^* = p_c(d + 1). \tag{B.13}$$

Under the default parameter setting $N = 50$, $r = 0.5$, $T = 2 \times 10^4$, $R_0 = 20$, $p_c = 50$, $\alpha = 0.5$, $\tau = 0.05$, and $\kappa = 1$, and using the homogeneous benchmark degree $d = \langle k \rangle = 4$, Eq. (B.11) yields

$$R^* = \frac{(\tau - 1)d + (p_c + \alpha - 1)(d + 1)}{\alpha + \tau} = \frac{(0.05 - 1) \cdot 4 + (50 + 0.5 - 1) \cdot 5}{0.5 + 0.05} \approx 443.09. \tag{B.14}$$

Moreover, the corresponding allocable resource per local pool is

$$G^* = p_c(d+1) = 50 \times 5 = 250. \tag{B.15}$$

Since

$$(1+r)e^* \approx 336.07 > 250, \tag{B.16}$$

the steady state indeed lies on the saturated branch, which justifies the equality case in the above derivation.

This result provides a homogeneous analytical benchmark for the steady-state resource level. We next extend the analysis to heterogeneous BA scale-free networks, where degree heterogeneity induces nontrivial resource inequality even under full cooperation.

Appendix B.2. Steady-state scale-free Gini analysis

We next analyze the resource inequality induced by degree heterogeneity under the all-cooperation steady state on a Barabási–Albert (BA) scale-free network. We assume that resources in each local pool are distributed equally among the members of the corresponding local group, and that the accumulated resource is at steady state.

Let individual i have degree d_i, and define

$$H_i := d_i + 1. \tag{B.17}$$

Then H_i is the size of the local group associated with individual i, including itself and its neighbours.

Under full cooperation, individual i contributes

$$e_i(t) = d_i + 1 + \alpha(R_i(t) - (d_i + 1)). \tag{B.18}$$

Moreover, in the high-resource regime $R_i(t) > d_i$, the consumption of individual i is

$$c_i(t) = d_i + \tau(R_i(t) - d_i). \tag{B.19}$$

For pool j, the capacity upper bound is

$$P_{\max,j}(t) = p_c\, N_{c,j}(t)\, \rho_{c,j}(t). \tag{B.20}$$

Under full cooperation, we have $N_{c,j}(t) = d_j + 1 = H_j$ and $\rho_{c,j}(t) = 1$, so that

$$P_{\max,j}(t) = p_c H_j. \tag{B.21}$$

Hence, on the saturated branch,

$$G_j^* = p_c H_j. \tag{B.22}$$

Since pool j distributes its allocable resource equally among the H_j members of its local group, each member receives

$$\frac{G_j^*}{H_j} = p_c \tag{B.23}$$

from that pool. Individual i belongs to exactly $H_i = d_i + 1$ pools, including its own pool and those of its neighbours. Therefore, its total received resource at steady state is

$$u_i^* = p_c H_i = p_c(d_i + 1). \tag{B.24}$$

At steady state, $R_i(t+1) = R_i(t) = R_i^*$, and thus

$$u_i^* = e_i^* + c_i^*. \tag{B.25}$$

Substituting the above expressions gives

$$p_c(d_i + 1) = d_i + 1 + \alpha(R_i^* - (d_i + 1)) + d_i + \tau(R_i^* - d_i). \tag{B.26}$$

After rearranging, we obtain

$$R_i^* = \frac{(\tau - 1)d_i + (p_c + \alpha - 1)(d_i + 1)}{\alpha + \tau}. \tag{B.27}$$

Equation (B.27) shows that the steady-state resource is an affine function of degree. Equivalently, by writing $H_i = d_i + 1$, we obtain

$$R_i^* = aH_i + b, \tag{B.28}$$

where

$$a = \frac{p_c + \alpha + \tau - 2}{\alpha + \tau}, \qquad b = \frac{1 - \tau}{\alpha + \tau}. \tag{B.29}$$

We now consider a BA scale-free network with attachment parameter $m = 2$. In the large-network limit, the degree distribution is

$$\mathbb{P}(d = k) = \frac{12}{k(k + 1)(k + 2)}, \qquad k \geq 2. \tag{B.30}$$

Hence, for $H = d + 1$,

$$\mathbb{P}(H = n) = \frac{12}{(n - 1)n(n + 1)}, \qquad n \geq 3. \tag{B.31}$$

For this distribution, one has

$$\mathbb{E}[H] = 5, \tag{B.32}$$

and the corresponding Gini coefficient is

$$G(H) = 24 - \frac{12}{5}\pi^2 \approx 0.31295. \tag{B.33}$$

Since R_i^* is an affine transform of H_i, the Gini coefficient of the steady-state resource distribution satisfies

$$G(R^*) = \frac{a\,\mathbb{E}[H]}{a\,\mathbb{E}[H] + b}\, G(H). \tag{B.34}$$

Substituting Eqs. (B.29) and (B.33), we obtain

$$G(R^*) = \frac{5(p_c + \alpha + \tau - 2)}{5(p_c + \alpha + \tau - 2) + (1 - \tau)} \left(24 - \frac{12}{5}\pi^2\right). \tag{B.35}$$

Under the default parameter setting $N = 50$, $r = 0.5$, $T = 2 \times 10^4$, $R_0 = 20$, $p_c = 50$, $\alpha = 0.5$, $\tau = 0.05$, and $\kappa = 1$, Eq. (B.35) gives

$$G_{BA}(R^*) = \frac{5(50 + 0.5 + 0.05 - 2)}{5(50 + 0.5 + 0.05 - 2) + (1 - 0.05)} \left(24 - \frac{12}{5}\pi^2\right) \approx 0.3117. \tag{B.36}$$

This value is derived from the asymptotic degree distribution of the BA network with $m = 2$, and therefore serves as a large-network theoretical benchmark. For a finite network such as $N = 50$, the observed Gini coefficient may deviate slightly from this value.

Therefore, under equal baseline and the all-cooperation steady state, the asymptotic theoretical Gini coefficient on a BA scale-free network with $m = 2$ is approximately 0.312.

Appendix B.3. Extension to ER and WS networks

The above BA analysis can be extended to other network topologies without repeating the full steady-state derivation. The key observation is that, under full cooperation, equal local redistribution, and the saturated-pool regime, the steady-state resource level is an affine function of the local group size $H_i = d_i + 1$, namely

$$R_i^* = aH_i + b, \tag{B.37}$$

where

$$a = \frac{p_c + \alpha + \tau - 2}{\alpha + \tau}, \qquad b = \frac{1 - \tau}{\alpha + \tau}. \tag{B.38}$$

Therefore, the ideal all-cooperation Gini coefficient can be written as

$$G(R^*) = \frac{a\bar{H}}{a\bar{H} + b} G(H), \tag{B.39}$$

where $\bar{H} = N^{-1} \sum_i H_i$ and $G(H)$ denotes the Gini coefficient of the local group size distribution.

For the ER network used in our simulations, we set $N = 50$ and $p_{\mathrm{ER}} = 4/(50 - 1)$. Thus, the expected degree is $\langle d \rangle = 4$, and the degree distribution is approximated by

$$d \sim \mathrm{Binomial}\left(49, \frac{4}{49}\right). \tag{B.40}$$

Substituting this distribution into the above formula gives

$$G_{\mathrm{ER}}(R^*) \approx 0.2122. \tag{B.41}$$

For the WS network, we set $N = 50$, $k = 4$, and rewiring probability $q = 0.1$. Since the WS degree distribution depends on the random rewiring realization, we use the corresponding expected degree-heterogeneity benchmark. This gives

$$G_{\mathrm{WS}}(R^*) \approx 0.0564. \tag{B.42}$$

These values show that the ideal all-cooperation Gini coefficient is mainly determined by the degree heterogeneity of the underlying topology. Even under persistent cooperation and equal local redistribution, global resource equality is not guaranteed in heterogeneous networks, because high-degree nodes participate in more local pools and therefore receive resources from more sources. Consequently, BA networks induce the largest degree-driven inequality, ER networks induce a moderate level of inequality, whereas WS networks with low rewiring probability remain close to the regular-network benchmark.

Appendix C. Additional Generalization Tests

To further examine the robustness of the learned and interpretable allocation mechanisms, we conduct additional generalization tests by varying key environmental parameters. Specifically, we change the pool-capacity coefficient p_c, the Fermi stochasticity parameter κ, and the pool growth factor r, while keeping the remaining settings unchanged. These tests evaluate whether the qualitative advantage of M2 persists beyond the default parameter configuration used in the main text.

It should be noted that changing p_c directly affects the scale of the average accumulated resource $\bar{R}(t)$, because p_c controls the upper capacity of local resource pools. Therefore, to make the results comparable across different parameter settings, we report the normalized average accumulated resource defined as

$$\bar{R}_{\mathrm{norm}}(t) = \frac{\bar{R}(t)}{R^*},$$

where R^* denotes the corresponding theoretical benchmark resource level under the given parameter setting in Appendix B. In addition, the resource-dependent binning used in M1 and M2 is also performed after this normalization. That is, the normalized resource level is first mapped into the corresponding resource bin, and then the allocation weights of the mixture mechanism are determined accordingly. This treatment ensures that the mixture mechanisms are compared under a unified resource scale when the environmental capacity changes.

Figures C.1–C.3 report the generalization results under variations of p_c, κ, and r, respectively. Overall, M2 exhibits stable performance across different parameter values. Its cooperation level, average accumulated resource, and Gini coefficient remain largely unchanged under these environmental shifts. By contrast, the original RL-Agent shows more visible performance fluctuations when the parameters deviate from the default setting. These results suggest that M2 does not merely overfit the default simulation environment, but captures a more robust allocation principle that generalizes across changes in resource capacity, stochasticity of strategy update, and pool growth intensity.

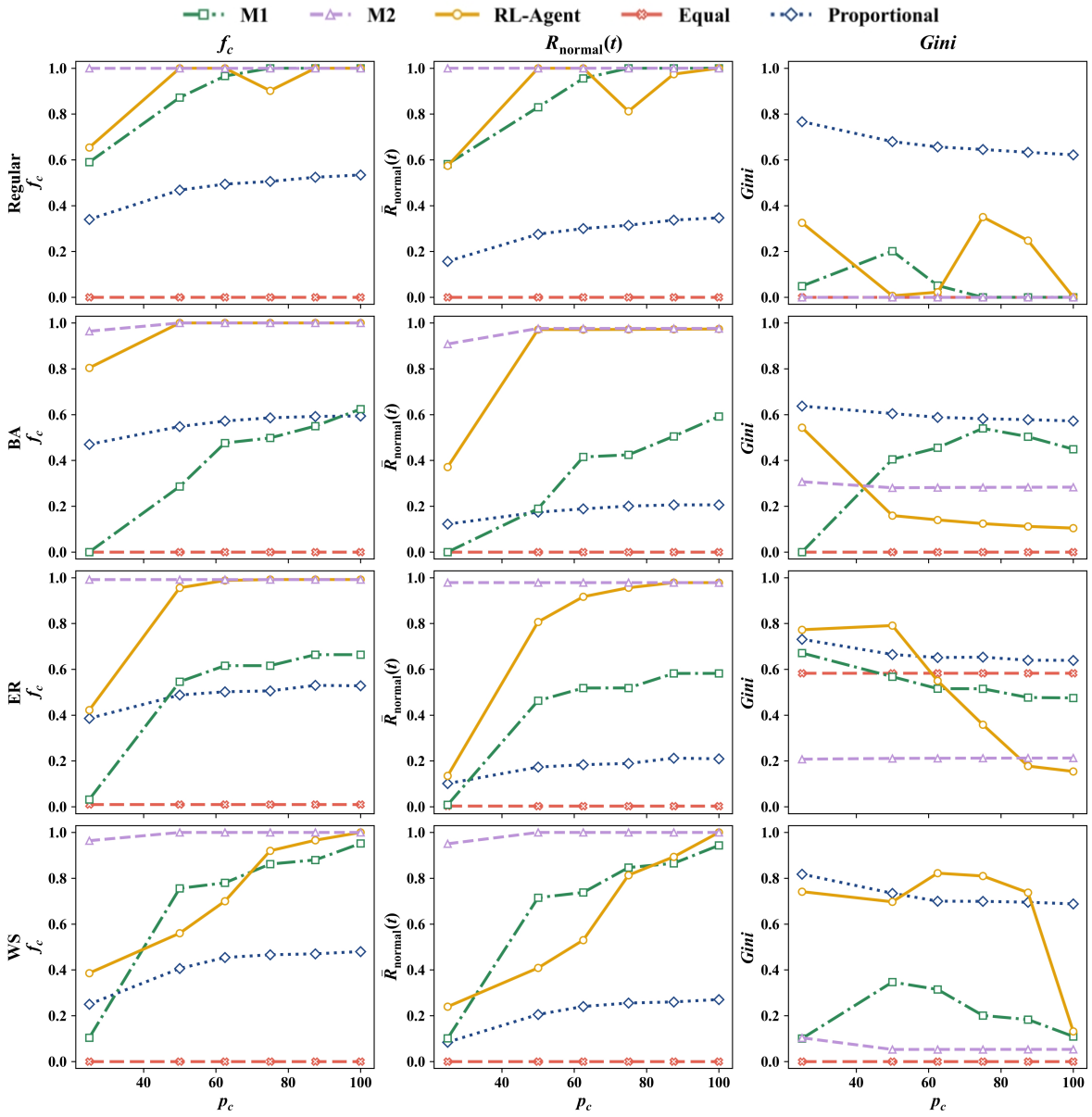

Figure C.1: Generalization test under different values of the pool-capacity coefficient p_c. The normalized average accumulated resource $\bar{R}_{\text{norm}}(t)$ is used to make results comparable across different resource-capacity scales. M2 maintains stable performance across different p_c values, whereas the original RL-Agent exhibits larger fluctuations.

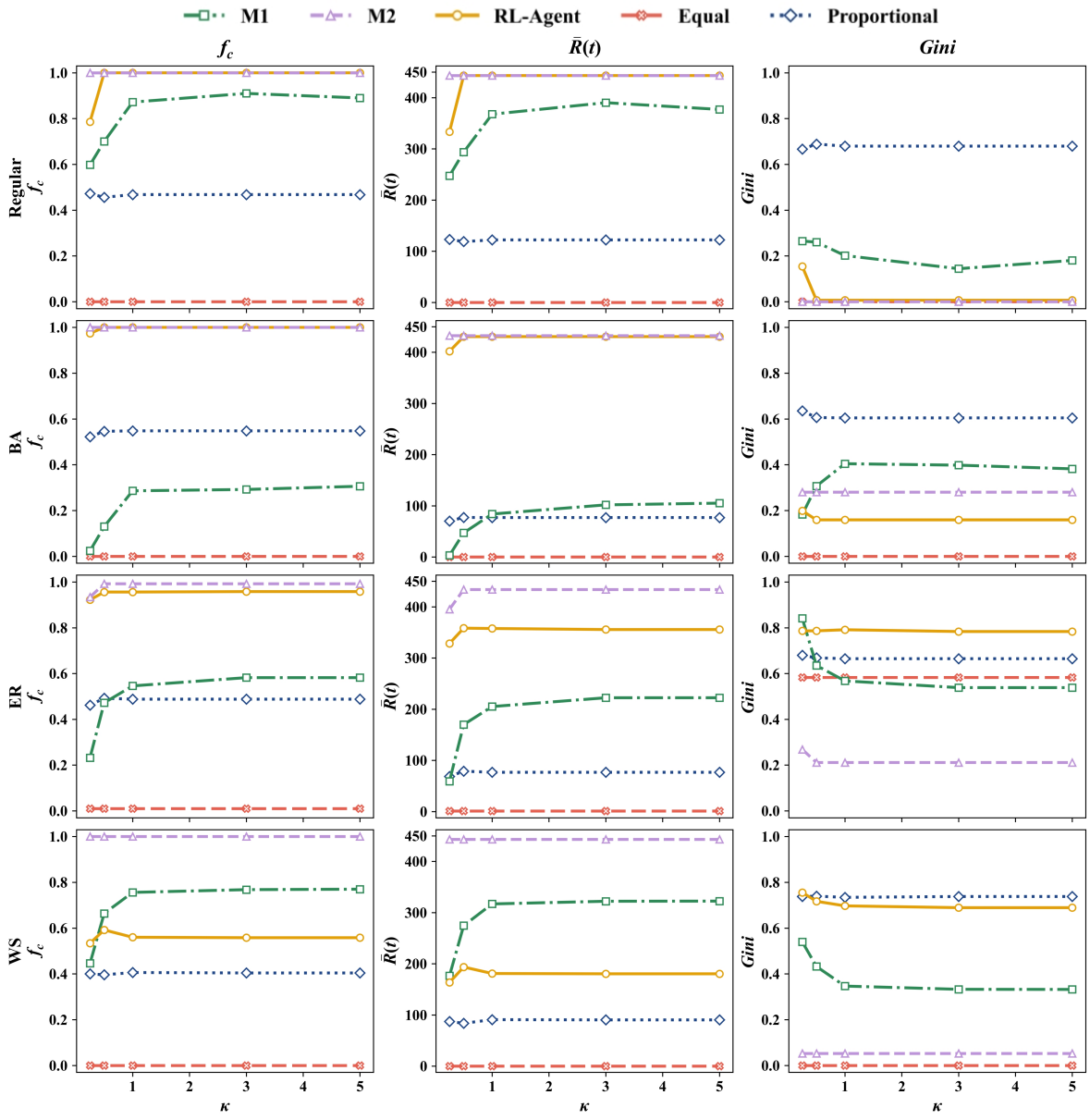

Figure C.2: Generalization test under different values of the Fermi stochasticity parameter κ. M2 remains robust when the stochasticity of strategy updating changes, while the original RL-Agent shows more visible sensitivity to the parameter shift.

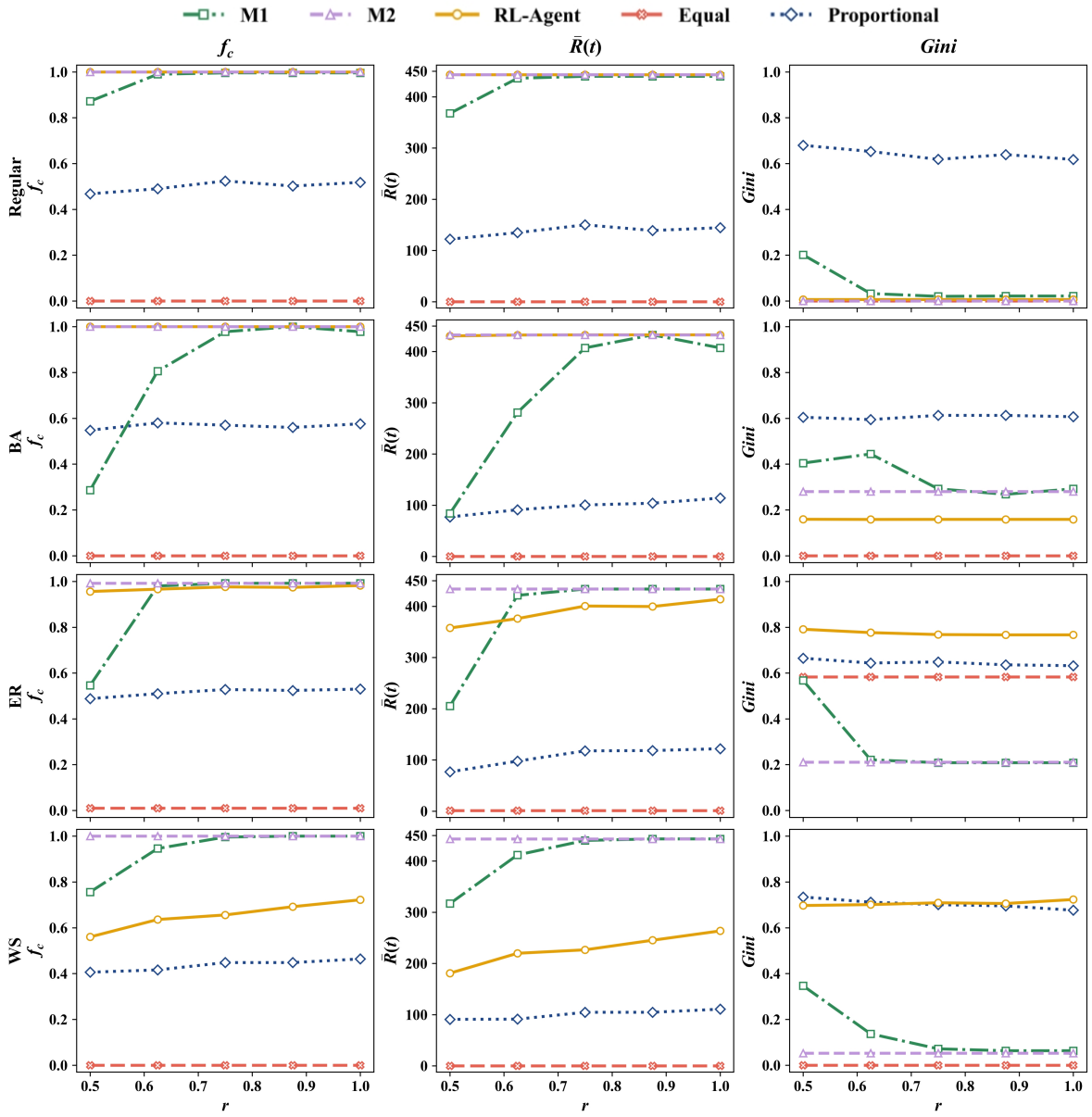

Figure C.3: Generalization test under different values of the pool growth factor r. M2 preserves its qualitative advantage across different resource-growth intensities, indicating that the extracted degree-conditioned mechanism generalizes beyond the default growth setting.

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